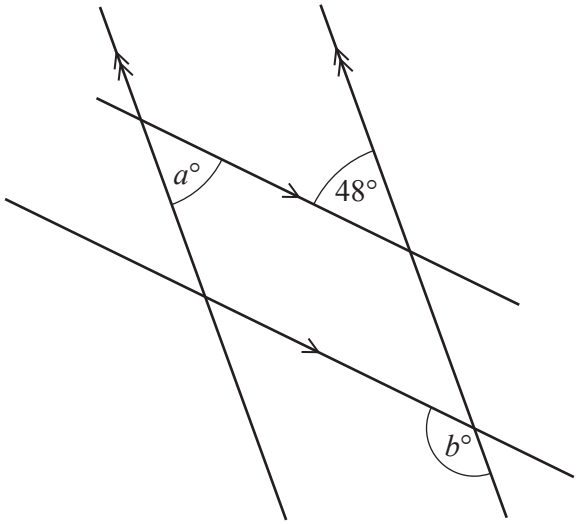
0580_w21_qp_31_Q4

4 **(a)**



NOT TO SCALE

The diagram shows two pairs of parallel lines.

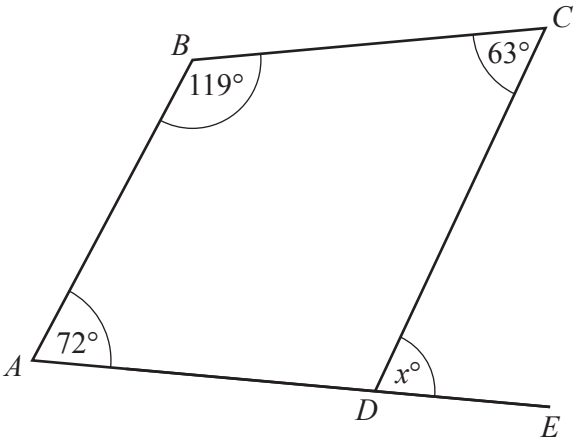
(i) Find the value of a.

$a =$ ..... [1]

(ii) Find the value of b.

$b =$ ..... [1]

(b)



NOT TO SCALE

The diagram shows a quadrilateral $ABCD$ and a straight line ADE.

Work out the value of x.

$x =$ ..... [2]

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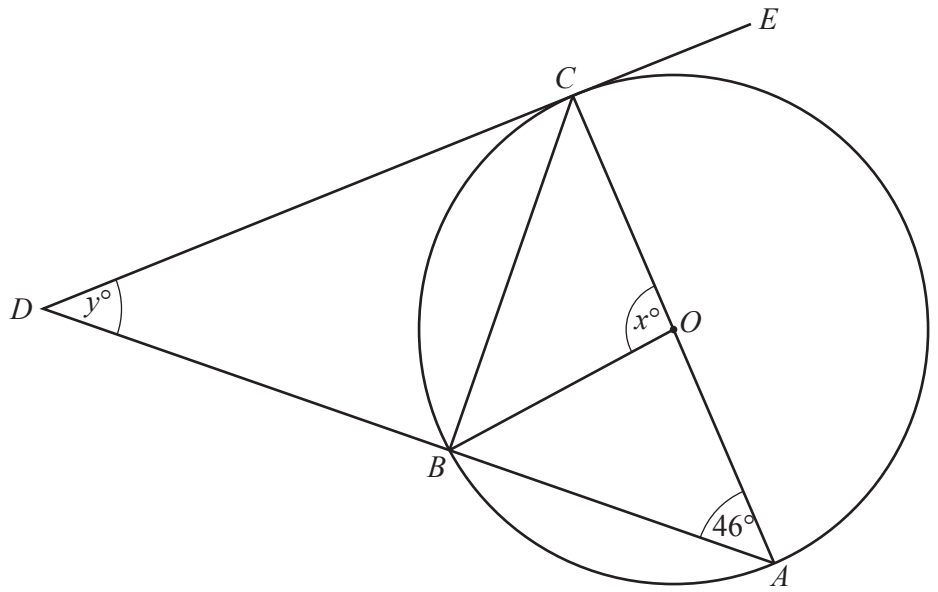
(c)



NOT TO SCALE

A, *B* and *C* are points on the circle, centre *O*.
AC is a diameter of the circle and *ABD* is a straight line.
DCE is a tangent to the circle at *C*.

(i) Write down the mathematical name for the line *BC*.

..... [1]

(ii) Explain why angle *ABC* is 90°.

..... [1]

(iii) Find the value of *x*.

x = ... [2]

(iv) Find the value of *y*.

y = ... [2]